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Submission of Original Research Article

Dear Editor-in-Chief,

We are pleased to submit our original research article, titled "*Temporal Structure Perception in Tonal and Atonal Music: An Online Behavioural Study*", for consideration for publication in the Annals of the New York Academy of Sciences. This work investigates a fundamental question in music cognition: how does tonal structure shape the perception of temporal information in music?

Across five online behavioural experiments ($N = 487$), we used a 'quilting' algorithm to rearrange segments of tonal (Bach) and atonal (Krenek) violin recordings across a wide range of temporal scales (60–3840 ms), and asked participants to rate perceived naturalness. Naturalness increased with segment length for both corpora, following a piecewise linear function with comparable slopes and elbow points, but a consistently higher plateau for tonal music, indicating that tonal and temporal structures contribute independently and additively to perception. A cross-domain comparison with speech revealed a reversed familiarity effect, suggesting domain-specificity in musical temporal processing. Musicians showed heightened sensitivity specifically in atonal music, while in non-musicians, self-reported Perceptual Abilities (Gold-MSI) robustly predicted the tonal elbow point and the tonal-atonal plateau difference, replicated via cross-validation.

These findings speak directly to the journal's interdisciplinary scope at the intersection of music psychology and auditory neuroscience. We confirm the manuscript is original, not under consideration elsewhere, and that all authors have approved the submission. The study was conducted in accordance with a protocol approved by the Duke University Institutional Review Board (IRB protocol number: D0206). Informed consent was obtained from all participants. The authors declare no conflicts of interest.

Best regards,

Dr. Seung-Goo Kim

On behalf of all co-authors.